

Abdoul Hamid COULIBALY

Backend Engineer — Data Systems & Cloud

📍 Rennes, France

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Software engineer working at the intersection of data, cloud infrastructure, and distributed systems. After several years building large-scale data processing architectures (IoT, telecom, renewable energy), I now focus on Infrastructure as Code components around Terraform, with a growing contribution to open source projects.

Professional Experience

Software Engineer – Open Source Terraform Provider

June 2026 - Present

INETUM (Orange Business) | Rennes, France

Cloud Avenue Terraform Provider

- Took over technical ownership of the Cloud Avenue Terraform provider, published on the Terraform Registry for platform customers.
- Ongoing development, maintenance, and improvement: Terraform resources and data sources, Go SDK, releases, and documentation.
- GitHub triage and support: user issues, feature requests, and guidance for external contributors.

Open Source Terraform Plugins

- Maintenance and evolution of 4 cross-provider Terraform plugins in the GitHub organization, used by external organizations including Microsoft.
- Open source community support: code reviews, issue triage, and contributor guidance.

Go Terraform HCL OpenAPI AI Agentic Development Git GitHub Actions CI/CD Open Source

Backend Engineer – Data Supply Chain

November 2023 - March 2026

Univers (QoS Energy) | La Chapelle-sur-Erdre, France

Architecture & Data Ingestion

- Development and refactoring of Go and Python microservices for ingesting millions of metrics per minute, in a GitOps-managed Kubernetes environment.
- Implementation and extension of event-driven architectures (RabbitMQ, NATS) for ingestion and asynchronous processing.
- Design of resilient pipelines handling network issues and source instability: distributed error handling, retry, backfilling and manual reload.

Integration & Connectors

- Development of connectors for heterogeneous sources: SCADA, REST, FTP, SQL databases.
- Active participation in architecture decisions: technology evaluation (NATS, Redis), data modeling and source API rate limiting management.
- Rewriting Go services to Python to align with Bazefield stack constraints.

Scalability & Resilience

- Service instrumentation with Prometheus and Grafana dashboard setup for operational monitoring.
- On-call rotation (1 week/6): critical incident resolution and L3 support for client technical operations.

Go Python Ruby RabbitMQ NATS PostgreSQL Kubernetes Azure GitOps Prometheus Grafana

Software Engineer

NumoData (formerly EXFO Solutions) | Rennes, France

Backend Development

Development of Go backend services for a network monitoring platform processing millions of metrics per minute.

Design of analytical queries (SQL, Druid): aggregations, multidimensional queries, optimization via indexing and caching.

Smart Usage Monitoring — Technical Ownership

End-to-end design and development of a dashboard usage analytics service: most viewed metrics, highest-volume dimensions, detection of underutilized data.

Results exposed via REST API, Prometheus instrumentation and Grafana dashboards, used for cache optimization and storage cost reduction.

Performance & Infrastructure

Load testing and correlation analysis of volume/resources (CPU, RAM, latency) for client sizing.

Development of an internal tool for automatic Kubernetes (Helm) configuration generation based on client sizing results.

Go

Python

Kafka

Apache Druid

AWS

Prometheus

Grafana

Helm

Kubernetes

Web Developer – Internship

Resalocal | Talloires-Montmin, France

May 2021 – August 2021

Development of an ingestion pipeline from the Apidae tourism database to the Akeneo PIM: import of POIs, events and activities.

Bidirectional synchronization: returning enriched data to Apidae after processing.

POI media management via Cloudinary API: image upload and transformation.

Handling ingestion constraints: pagination, rate limiting, and preservation of existing PIM data.

PHP

Laravel

Docker

Kubernetes

Akeneo

Cloudinary

REST API

Technical Skills

</> Languages

Go Python Ruby TypeScript SQL Bash

Cloud & DevOps

AWS Azure Docker Kubernetes Terraform Helm GitLab CI

Data & Messaging

PostgreSQL MySQL Apache Druid Kafka RabbitMQ NATS

Observability

Prometheus Grafana OpenTelemetry

Education

Master's in Computer Science – DABI (Decision-Making, Learning & Big Data) | University of Rennes 12021 – 2023

Data warehousing, data mining, Cloud & Big Data, artificial intelligence, application integration

Bachelor's in Computer Science – IT Management | University of Rennes 12020 – 2021

IT/management dual competency, databases, web development, object modeling, data analysis

Higher Technician Diploma in Computer Science | INP-HB2016 – 2019

Algorithms, networking, web development, system administration, project management

General & Professional English Training | Wall Street English2024

Oral and written communication, IT technical vocabulary, professional presentations, meetings in English

Languages

French (native)

English (fluent)

Bambara (native)